Symmetry

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1 Motivation

Topic: symmetry. On the way we collect several elementary techniques of differential geometry that are urgently needed in Einstein's theory (and some loose ends from earlier lectures).

The intuition: the round sphere (S^2, g_{round}) has rotational symmetry, while the "potato" does not. As *smooth manifolds* both are S^2; a smooth manifold alone has no shape. What distinguishes them is the *metric*: round metric versus "potato metric". Hence

> symmetry is a property of the *metric*, not of the manifold.

This differs from the way rotations were met before, as the group $SO(3)$ of linear maps A on a vector space with an inner product, $\langle Ax, Ay\rangle = \langle x, y\rangle$. A metric gives an inner product on every tangent space, but there are many inner products and many tangent spaces; the question is whether the *distribution* of these inner products over the manifold is invariant under rotations. So we need new tools. The question is: *how do we describe symmetries of a metric?*

This is not academic: nobody can solve Einstein's equations without further assumptions such as symmetry of the solution. Textbooks often say "make the following ansatz for a spherically symmetric metric" and then handwave; here we learn how to do it properly.

2 The push-forward

Let M, N be smooth manifolds (no metric yet) and $\phi : M \to N$ a smooth map.

Definition 1 (Push-forward)**.** The *push-forward* $\phi_* : TM \to TN$ takes a vector $X \in T_pM$ to $\phi_* X \in T_{\phi(p)}N$, defined by its action on smooth functions $f : N \to \mathbb{R}$:

$$(\phi_* X)(f) := X(f \circ \phi).$$

The base point p is mapped to $\phi(p)$, so the fibre T_pM is sent into the fibre $T_{\phi(p)}N$ (this is a linear map between these two spaces). *Mnemonic: vectors are pushed forward.*

Components. Choose charts (U, x) of M and (V, y) of N. The components of ϕ_* are

$$(\phi_*)^a{}_i := \mathrm{d}y^a\big(\phi_*\, \partial/\partial x^i\big) = \frac{\partial}{\partial x^i}(y^a \circ \phi) = \frac{\partial \hat{\phi}^a}{\partial x^i},$$

where $\hat{\phi} = y \circ \phi \circ x^{-1}$ is the chart representation of ϕ. Here $i = 1, \ldots, \dim M$ and $a = 1, \ldots, \dim N$, so these are *not* tensor components: they refer to different vector spaces, in different manifolds. They are just the Jacobian matrix of $\hat{\phi}$.

2.1 Tangent vectors of curves

Theorem 1. *Let $\gamma : I \to M$ be a smooth curve through $p = \gamma(\lambda_0)$. Then*

$$\phi_*\big(v_{\gamma,p}\big) = v_{\phi\circ\gamma,\,\phi(p)},$$

i.e. the push-forward of the tangent vector of a curve is the tangent vector of the pushed-forward curve $\phi \circ \gamma$.

Proof. For any $f \in C^\infty(N)$,

$$\big(\phi_* v_{\gamma,p}\big)(f) = v_{\gamma,p}(f \circ \phi) = (f \circ \phi \circ \gamma)'(\lambda_0) = v_{\phi\circ\gamma,\phi(p)}(f),$$

using the definition of ϕ_*, then the definition of the tangent vector of a curve, then associativity of composition. As this holds for every f, the claim follows. □

Example 1 (Embeddings: intrinsic versus extrinsic vectors)**.** If $\dim M < \dim N$ and ϕ is an embedding (smooth and injective), say $M = \mathbb{R}^2$ placed in $N = \mathbb{R}^3$ as a bent sheet, then ϕ_* turns an *intrinsic* tangent vector (the velocity an ant on the surface, knowing nothing about the third dimension, would report) into the *extrinsic* one (the velocity we see in $\mathbb{R}^3$). Same object, two perspectives. (Another application, with $M = N$, follows below.)

3 The pull-back

Definition 2 (Pull-back)**.** For a smooth map $\phi : M \to N$, the *pull-back* $\phi^* : T^*N \to T^*M$ takes a covector $\omega \in T^*_{\phi(p)}N$ to $\phi^*\omega \in T^*_pM$,

$$(\phi^*\omega)(X) := \omega(\phi_* X), \qquad X \in T_pM.$$

Note the direction: the pull-back goes from the cotangent bundle of the *target* to that of the *domain*, but it is *defined using the push-forward. Mnemonic: vectors are pushed forward, covectors are pulled back.* (Index up ϕ^* for the pull-back, index down ϕ_* for the push-forward.)

Components. With the same charts,

$$(\phi^*)_i{}^a = (\phi^*\mathrm{d}y^a)\Big(\frac{\partial}{\partial x^i}\Big) = \mathrm{d}y^a\Big(\phi_*\frac{\partial}{\partial x^i}\Big) = (\phi_*)^a{}_i.$$

The components of pull-back and push-forward are *numerically the same* (the Jacobian); one calculates them once, and which map one gets depends on which index is contracted: with the lower index i one pushes a vector forward, with the upper index a one pulls a covector back.

Picture. The gradient of a function f on N (a covector, with level sets like isotherms) is pulled back to the gradient of $f \circ \phi$ on M.

Remark. Push-forward and pull-back are *not* inverses of each other: they act on four different spaces, in opposite directions. In linear algebra language, the pull-back is the dual (adjoint) map of the push-forward. Generically a vector cannot be pulled back and a covector cannot be pushed forward; only if ϕ is invertible can one do both, using ϕ^{-1}. These constructions act on the fibre over each point (they are not about mapping fields to fields so that a section stays a section).

4 The induced metric

Let $\phi : M \to N$ be an embedding and let N carry a metric g_N. The *induced metric* on M is the pull-back

$$g_M := \phi^* g_N, \qquad (g_M)_p(X, Y) := (g_N)_{\phi(p)}\big(\phi_* X, \phi_* Y\big).$$

In components, with ϕ_* as above,

$$(g_M)_{ij}(p) = (g_N)_{ab}(\phi(p))\, \frac{\partial \hat{\phi}^a}{\partial x^i}(p)\, \frac{\partial \hat{\phi}^b}{\partial x^j}(p).$$

Intuition: lengths of curves in M are measured as the lengths of their images in the ambient space, as a string laid on the surface.

Example 2. Take $N = \mathbb{R}^3$ with the Euclidean metric (δ_{ab} in the identity chart) and $M = S^2$.

- With the standard embedding as the unit sphere, $\phi^* g_{\text{Eucl}} = g_{\text{round}}$: this is where the round metric used earlier comes from.
- With an embedding as an ellipsoid, one obtains a different metric on the *same* manifold S^2.

The embedding matters for the result: *the shape lies in the metric.* (The inverse metric can analogously be induced by pulling back covectors.)

5 Integral curves, completeness and flows

Definition 3 (Integral curve)**.** Let X be a vector field on M. A curve $\gamma : I \to M$, $I \subseteq \mathbb{R}$ an interval, is an *integral curve* of X if its tangent vector reproduces the vector field at each point of the curve:

$$v_{\gamma,\gamma(\lambda)} = X_{\gamma(\lambda)} \qquad \text{for all } \lambda \in I.$$

Picture: X is the velocity field of the water molecules in a river, and an integral curve is the track of a paper boat with no propulsion.

Definition 4 (Complete vector field)**.** X is *complete* if every integral curve can be defined on all of $I = \mathbb{R}$.

Example 3. A vector field on a surface (with a stationary point where $X = 0$): its integral curves can be extended to all of $\mathbb{R}$ (the one through the zero being the constant curve). If one *removes* that point from the manifold (not from the field!), the same field becomes incomplete: the integral curves heading towards the missing point run out at a finite parameter value.

One cannot repair this by reparametrizing: the field X fixes the parametrization (the velocity must be exactly X, not twice or half). In spacetime physics, *singularities* are points removed from the manifold, e.g. because the curvature blows up; roughly, singularities are points where curves end at a finite parameter value, which is the technical door to the Hawking–Penrose singularity theorems. Completeness is therefore an important notion.

Theorem 2. *A smooth vector field with compact support is complete. (In particular every smooth vector field on a compact manifold is complete.)*

Definition 5 (Flow)**.** The *flow* of a complete vector field X is the map

$$h^X : \mathbb{R} \times M \to M, \qquad (\lambda, p) \mapsto h^X(\lambda, p) := \gamma_p(\lambda),$$

where $\gamma_p : \mathbb{R} \to M$ is the integral curve of X with $\gamma_p(0) = p$. By convention one writes the first argument as a subscript, $h^X_\lambda(p)$.

For fixed λ, $h^X_\lambda : M \to M$ is a smooth map that pushes every point a parameter distance λ along the integral curves of X; a set of points is carried by the river and generally evolves into a different set. (One never steps into the same river twice.) No metric has been used: this is defined on a bare smooth manifold. Since h^X_λ is a map from M to itself, the push-forward and pull-back of Sections 2 and 3 apply to it with $M = N$.

6 The Lie algebra of vector fields

Let $\Gamma(TM)$ denote the vector fields on M. It is a $C^\infty(M)$-module, and also, restricting scalars to $\mathbb{R}$, an (infinite dimensional) real vector space. On it there is the commutator

$$[X, Y](f) := X(Y(f)) - Y(X(f)),$$

which is again a vector field and satisfies

(i) $[X, Y] = -[Y, X]$ (antisymmetric),

(ii) $[\lambda X + Z, Y] = \lambda[X, Y] + [Z, Y]$ for $\lambda \in \mathbb{R}$ ($\mathbb{R}$-*linear*),

(iii) $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$ (Jacobi identity).

Any vector space with such a bracket is called a *Lie algebra*; so $\big(\Gamma(TM), [\cdot, \cdot]\big)$ is a (huge) Lie algebra.

Definition 6 (Lie subalgebra, structure constants)**.** Let $X_1, \dots, X_s$ be vector fields on M such that

$$[X_i, X_j] = C^k{}_{ij}\, X_k, \qquad C^k{}_{ij} \in \mathbb{R} \text{ (real numbers, not functions)},$$

for all $i, j = 1, \dots, s$. The $C^k{}_{ij}$ are the *structure constants*. Then the *real* span $L = \operatorname{span}_{\mathbb{R}}\{X_1, \dots, X_s\}$ is closed under the bracket and is an s-dimensional *Lie subalgebra* of $\Gamma(TM)$.

There are many choices of such subalgebras.

Example 4 ($\mathfrak{so}(3)$ on S^2)**.** On S^2, in the (θ, φ) chart (which covers almost all of the sphere), take

$$\begin{aligned} X_1 &= -\sin\varphi\,\frac{\partial}{\partial\theta} - \cot\theta\cos\varphi\,\frac{\partial}{\partial\varphi}, \\ X_2 &= \;\;\cos\varphi\,\frac{\partial}{\partial\theta} - \cot\theta\sin\varphi\,\frac{\partial}{\partial\varphi}, \\ X_3 &= \frac{\partial}{\partial\varphi}. \end{aligned}$$

A direct calculation gives $[X_i, X_j] = -\epsilon_{ijk} X_k$ (the sign depends on convention, and the missing i compared with quantum mechanics's $[L_i, L_j] = \mathrm{i}\epsilon_{ijk} L_k$ is absorbed by working with real numbers). This is the angular momentum or rotation algebra, $\mathfrak{so}(3)$. Remarkably, there is *no metric anywhere*: these are three vector fields on the merely smooth manifold S^2. To be proper one must also give the same fields in a second chart, compatible on the overlap. The flows of these fields are the rotations.

7 Symmetry of a metric

Definition 7 (Symmetry)**.** Let g be a metric tensor field on M. A finite dimensional Lie subalgebra $L \subseteq \Gamma(TM)$ is said to be a *symmetry* of g if for every (complete) vector field $X \in L$ and every $\lambda \in \mathbb{R}$,

$$\big(h^X_\lambda\big)^* g = g,$$

i.e. for all $p \in M$ and $A, B \in T_pM$,

$$g_{h_\lambda^X(p)}\big((h_\lambda^X)_* A,\ (h_\lambda^X)_* B\big) = g_p(A, B).$$

Here X ranges over the whole real span of $X_1, \dots, X_s$ (not the C^∞ span). In words: flowing along any field of L changes nothing about the metric. The maps h_λ^X are *isometries*, so each $X \in L$ generates a one-parameter family of isometries. (That the vector fields in L are complete is guaranteed in the situation where L comes from the Lie algebra of a Lie group acting on M; this was only mentioned, not proved.)

Remark (A correction made in the lecture). The lecturer first wrote the condition with the wrong arguments; the correct statement is the pull-back identity above, where the vectors A, B are arbitrary tangent vectors at a point, and the vector field X enters only through the flow.

Example 5. Let $L = \operatorname{span}_{\mathbb{R}}\{X_1, X_2, X_3\}$ be the $\mathfrak{so}(3)$ above.

- L is a symmetry of the round metric g_{round} (the sphere is rotationally symmetric).
- It is *not* a symmetry of the potato metric.
- For an ellipsoid with axes $a = b \neq c$ (it looks circular from above), only the one-dimensional subalgebra $\operatorname{span}\{X_3\}$ (rotations about the c-axis) is a symmetry: a cylindrical symmetry, with trivial structure constants.

This is the sense in which the round sphere "is $SO(3)$ symmetric" while the ellipsoid is not.

Application. In spacetime the metric will be Lorentzian, and "spherically symmetric" means exactly this condition applied to the four-dimensional metric. Likewise cosmology: with no metric yet, one demands a six-dimensional symmetry algebra (three translations and three rotations), giving intuitively homogeneity and isotropy.

8 The Lie derivative

The definition of symmetry is geometrically intuitive but hard to check: one needs all λ and all fields in L. Subtract g, divide by λ and let $\lambda \to 0$: if $(h_\lambda^X)^* g = g$ for all λ, then

$$\lim_{\lambda \to 0} \frac{1}{\lambda}\Big[(h_\lambda^X)^* g - g\Big] = 0.$$

The left-hand side is the *Lie derivative* $\mathcal{L}_X g$ (this identification is quoted, not proved, in the lecture). So L is a symmetry of g if $\mathcal{L}_X g = 0$ for all $X \in L$, and this is easy to check.

Definition 8 (Lie derivative)**.** On a smooth manifold (no extra structure needed), the *Lie derivative* is the map sending a vector field X and a (p, q) tensor field T to a (p, q) tensor field $\mathcal{L}_X T$ such that

(1) $\mathcal{L}_X f = X(f)$ for functions f;

(2) $\mathcal{L}_X Y = [X, Y]$ for vector fields Y;

(3) $\mathcal{L}_X(S + T) = \mathcal{L}_X S + \mathcal{L}_X T$;

(4) Leibniz rule, e.g. for a $(1, 1)$ tensor, $\mathcal{L}_X\big(T(\omega, Y)\big) = (\mathcal{L}_X T)(\omega, Y) + T(\mathcal{L}_X \omega, Y) + T(\omega, \mathcal{L}_X Y)$, and similarly for any valence;

(5) $\mathcal{L}_{X+Z} T = \mathcal{L}_X T + \mathcal{L}_Z T$ (additive in X).

This looks like the definition of the covariant derivative, but is weaker: the covariant derivative is required to be C^∞-linear in the lower slot ($\nabla_{fX} = f\nabla_X$), while here only additivity is required. Axiom (2) is what fixes everything, so no extra structure is needed.

Warning. $\mathcal{L}_X$ is *not* C^∞-linear in X: $\mathcal{L}_{fX}Y = [fX, Y] = f[X, Y] - Y(f)\,X$. The price for not imposing extra structure is that one must prescribe the vector field X in a neighbourhood, not just the vector at the point: the derivative uses the flow, hence information about how X behaves nearby. (One does *not* need X to be complete to define $\mathcal{L}_X$, only to be a vector field near the point.)

Components. In a chart, for a vector field Y and a $(1,1)$ tensor T:

$$(\mathcal{L}_X Y)^i = X^m \partial_m Y^i - Y^m \partial_m X^i,$$
$$(\mathcal{L}_X T)^i{}_j = X^m \partial_m T^i{}_j - (\partial_m X^i)\, T^m{}_j + (\partial_j X^m)\, T^i{}_m.$$

Compare with the covariant derivative: the first term is the same, but each upper index comes with a *minus* (the covariant derivative has a plus), each lower index with a plus, and instead of the connection coefficients the *derivatives of* X appear (and so X does not factor out). For the metric g, with two lower indices,

$$(\mathcal{L}_X g)_{ij} = X^m \partial_m g_{ij} + (\partial_i X^m)\, g_{mj} + (\partial_j X^m)\, g_{im}.$$

Remark (Use in practice). It is now straightforward to check whether a metric has a given symmetry: compute $\mathcal{L}_X g$ in a chart for each generator. Conversely, take the most general metric, impose $\mathcal{L}_X g = 0$, and see how this restricts its components in that coordinate system. This gives a legitimate ansatz for the Einstein equations with many components already zero, with no handwaving. One only has to say which symmetry algebra one wants implemented.

Questions raised.

- Is the algebra more fundamental than the group? In practice one specifies the vector fields (and their commutators) from the physical idea of the symmetry; the translation algebra, for instance, has all structure constants zero and is not semisimple.
- Subtraction and division in the limit are carried out in the vector space of $(0,2)$ tensor fields (addition and scalar multiplication there).

Summary

- Symmetry is a property of a metric, not of a smooth manifold: round sphere versus potato, both S^2.
- Push-forward ϕ_* (vectors, along ϕ; components = Jacobian) and pull-back ϕ^* (covectors, against ϕ; defined via ϕ_*; same Jacobian components). The tangent vector of a curve is pushed to the tangent vector of the image curve.
- Induced metric $g_M = \phi^* g_N$ for an embedding: the shape is in the metric (round sphere from $\mathbb{R}^3$, a different one for an ellipsoid).
- Integral curves, completeness (compact support implies complete), flow h^X_λ of a complete vector field.
- Vector fields form a Lie algebra under the commutator; a finite dimensional subalgebra with real structure constants, e.g. $\mathfrak{so}(3)$ on S^2, requires no metric.

- L is a symmetry of g iff $(h^X_\lambda)^* g = g$ for all $X \in L$ and all λ; the h^X_λ are isometries.
- Equivalently $\mathcal{L}_X g = 0$ for all $X \in L$, easily computed in a chart: $(\mathcal{L}_X g)_{ij} = X^m \partial_m g_{ij} + \partial_i X^m g_{mj} + \partial_j X^m g_{im}$.